

Single-Cell RNA-seq Subsetting Analysis of Intestinal IEL and LPL Compartments: Transcriptomic Refinement and Cell-Type Resolution

Bioinformatics Analysis Service 2026-08-04

1. Executive Summary

This report describes the compartment-specific subsetting analysis performed on the integrated single-cell RNA-seq atlas of the human intestinal immune system.

Following the global transcriptomic profiling and consensus cell-type annotation, the dataset was stratified into two major functional immune compartments:

Intraepithelial lymphocytes (IEL) Lamina propria lymphocytes (LPL)

The objective of this analysis was to resolve compartment-specific transcriptional programs, refine cellular heterogeneity within each immune niche, and enable downstream differential and functional interpretation at increased biological resolution.

The analytical workflow included:

- extraction of IEL and LPL subsets from the globally integrated Seurat object
- re-evaluation of quality and distribution of cells per compartment
- compartment-specific normalization and dimensionality reduction
- visualization of transcriptional structure within each compartment
- reassessment of cluster structure in compartmental space
- validation of compartment identity using canonical markers
- preparation of refined objects for downstream differential and enrichment analyses

This subsetting step represents the transition from a system-level atlas to a niche-resolved immune landscape.

2. Biological Rationale for Subsetting

The intestinal immune system is organized into spatially and functionally distinct compartments, each characterized by specific lymphocyte populations and transcriptional programs.

The separation into IEL and LPL compartments is biologically motivated by:

IEL compartment

- enrichment for tissue-resident cytotoxic T cells
- presence of $\gamma\delta$ T cells and unconventional T cell subsets
- strong epithelial interaction signatures

LPL compartment

- enrichment for helper T cells, T follicular-like populations, and regulatory T cells
- higher immunomodulatory and cytokine-driven transcriptional programs
- stronger antigen presentation and adaptive immune signaling

This separation enables the resolution of transcriptional heterogeneity that is masked in the global integrated analysis.

3. Data Source and Input Object

The subsetting analysis was performed on the globally integrated Seurat object obtained from the full intestinal atlas.

This object contains:

- Harmony-integrated latent space
- Consensus cell-type annotations
- Global UMAP embedding
- Graph-based cluster assignments
- Compartment annotation (IEL / LPL)

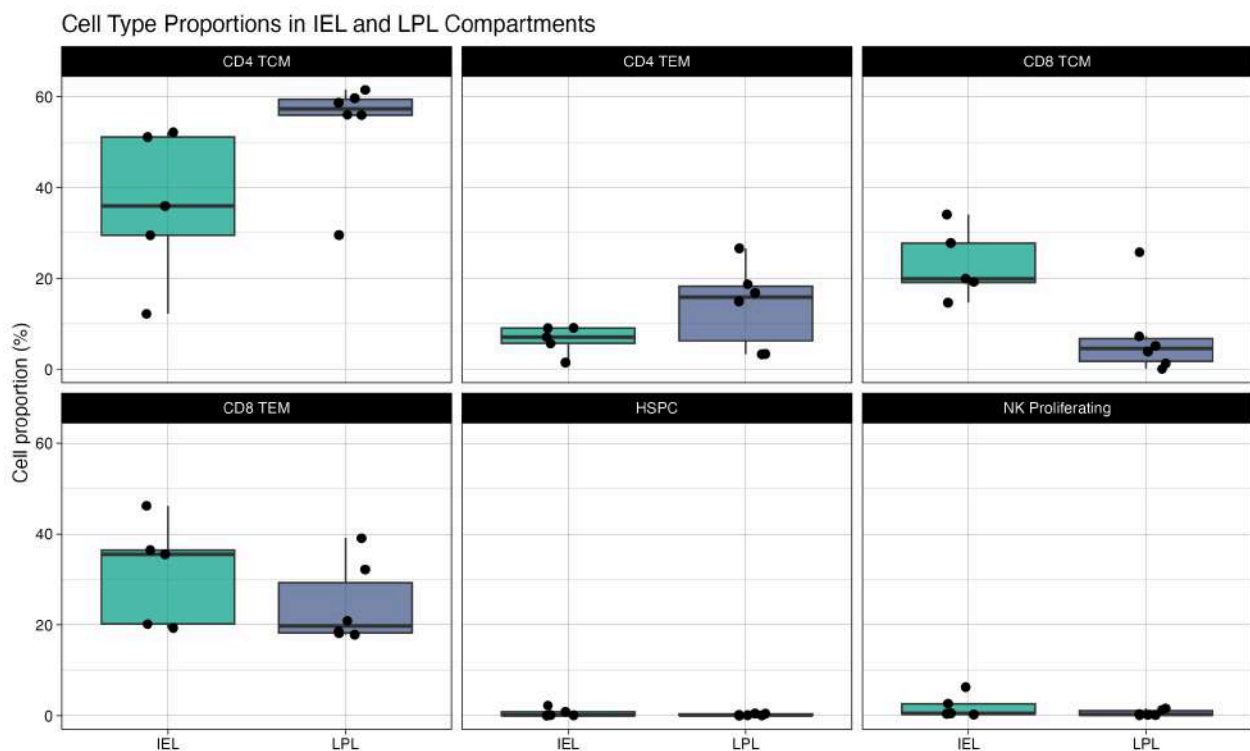
4. Compartment Stratification

Cells were stratified into two major immune compartments based on annotation-derived labels:

- Intraepithelial lymphocytes (IEL)
- Lamina propria lymphocytes (LPL)

This stratification was followed by compartment-specific reclustering to resolve internal transcriptional heterogeneity.

4.1 Cell distribution across compartments



5. Compartment-Specific Reclustering Strategy

Following extraction of IEL and LPL subsets, each compartment was independently reanalyzed to resolve internal transcriptional heterogeneity.

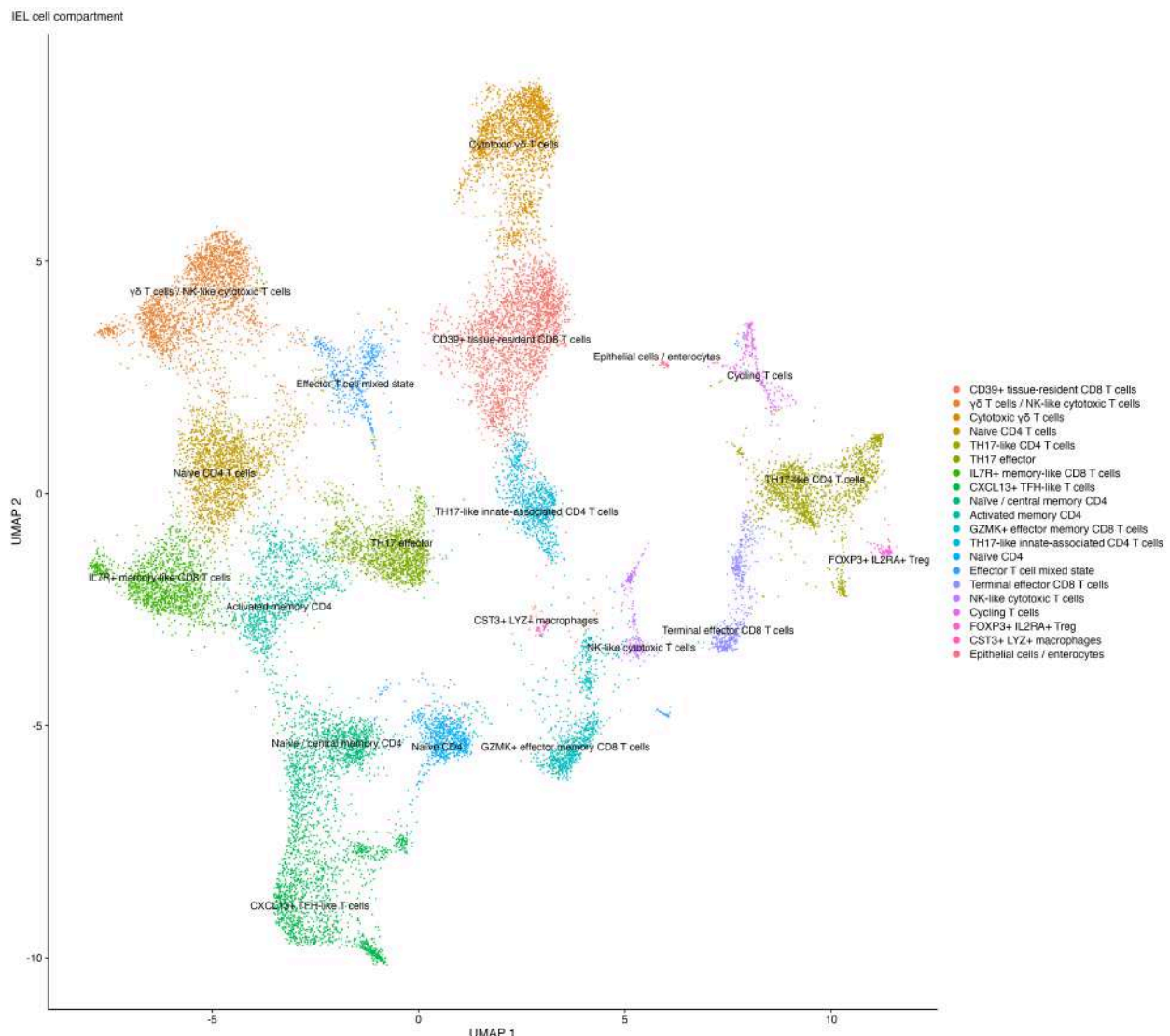
This step involved:

- reclustering within each compartment
- evaluation of transcriptional substructure
- mapping of clusters to biologically interpretable cell types

This approach allows resolution of fine-grained immune states that are masked in global clustering.

6. IEL Compartment Analysis and Reclustering

6.1 IEL transcriptional landscape



The IEL compartment reveals a highly structured and transcriptionally heterogeneous landscape characterized by tissue-resident and cytotoxic programs, with additional contributions from γδ T cell, TH17-like, and regulatory T cell states.

6.2 IEL clustering and cell-type consolidation

Reclustering of IEL cells identified:

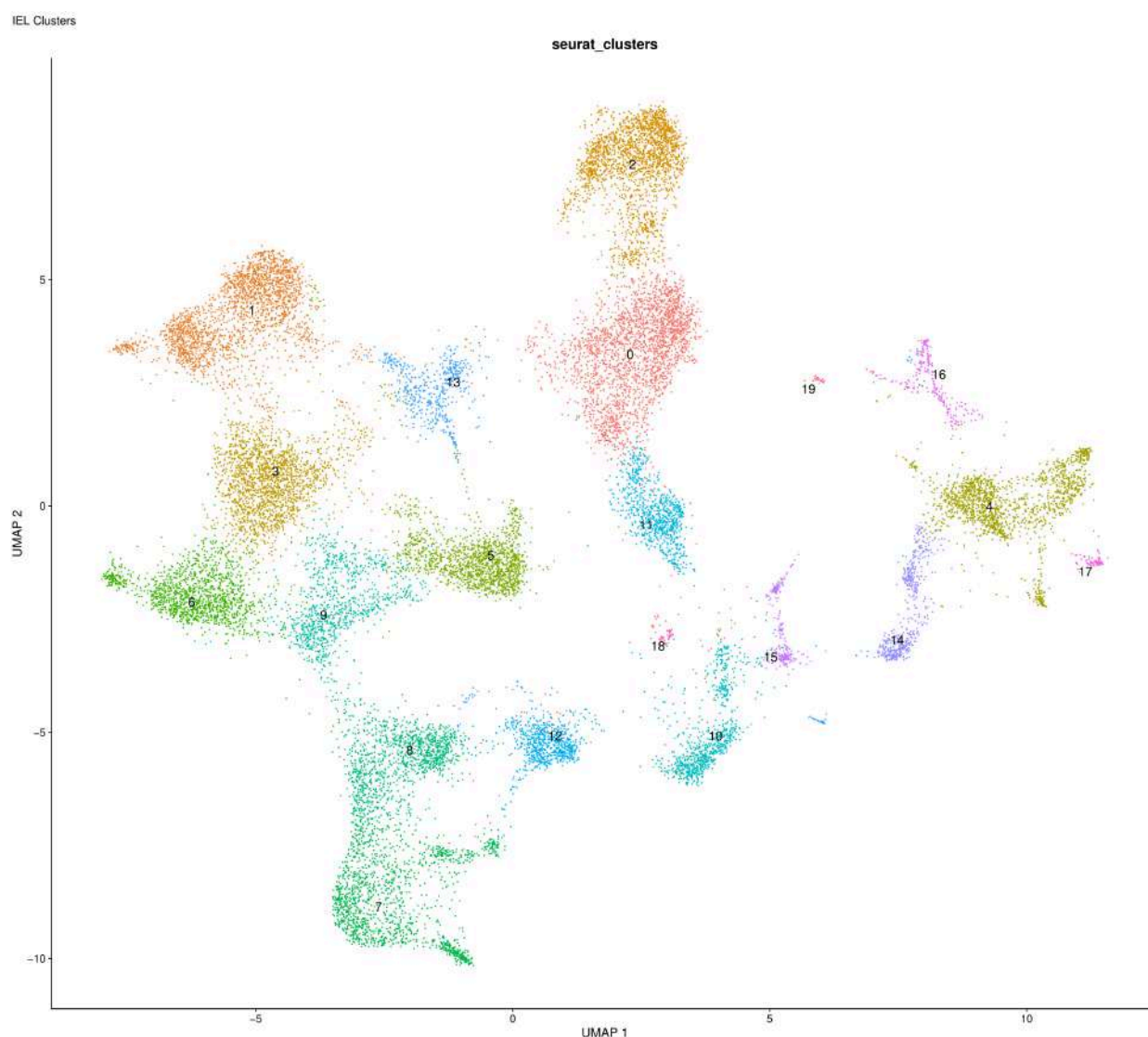
- 20 transcriptionally distinct clusters

These clusters were consolidated into:

- 20 biologically annotated cell types

This near 1:1 mapping indicates a high-resolution transcriptional fragmentation of the IEL compartment, reflecting strong cellular plasticity within the epithelial immune niche.

Clusters IEL Compartment



6.3 Key marker expression in IEL

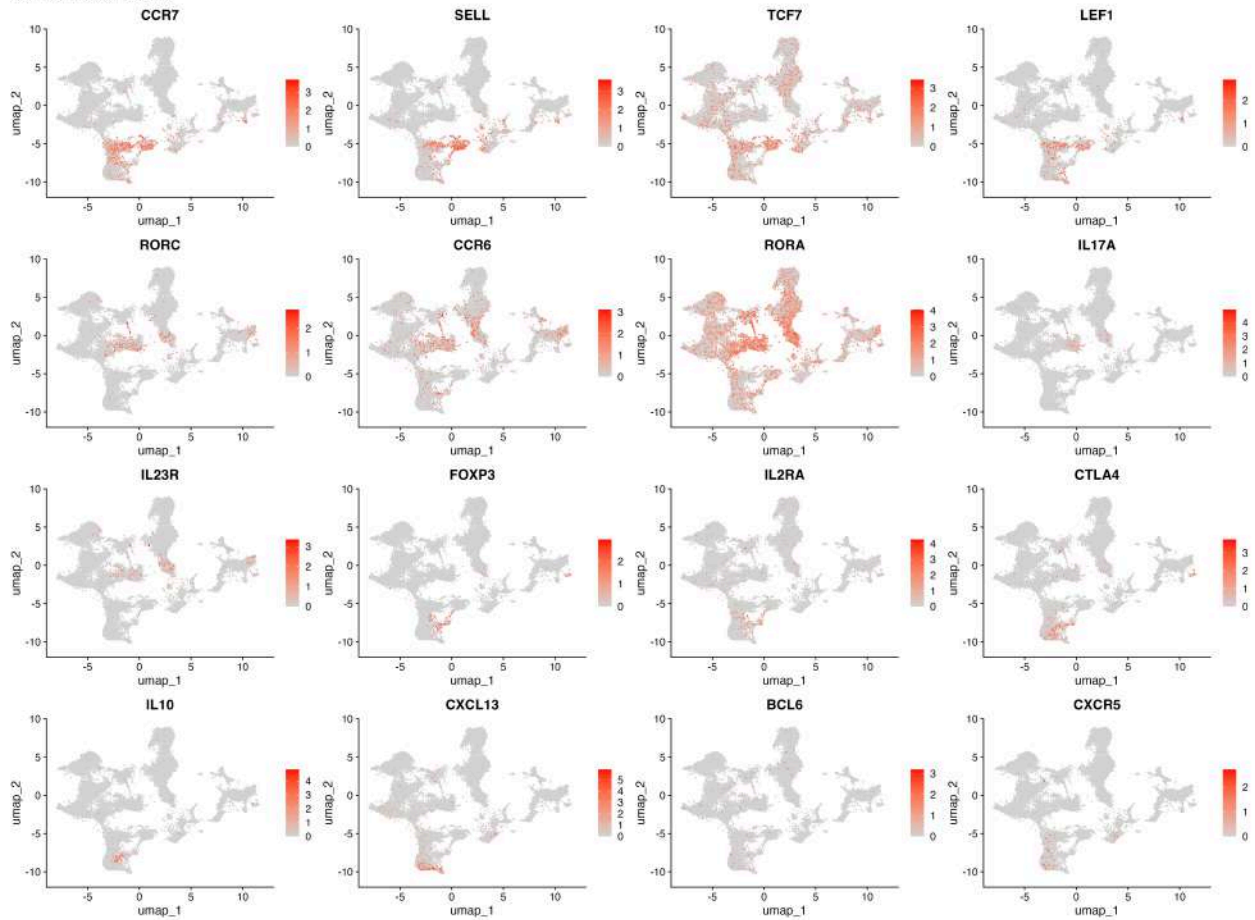
Canonical markers were used to validate IEL identity and substructure, including:

- Cytotoxic signature genes (e.g. GZMB, PRF1, NKG7)
- Tissue residency markers (e.g. CD69, ITGAE)
- $\gamma\delta$ T cell markers (e.g. TRDC, TRGC1/2)
- TH17-like T cell markers (e.g. RORA, IL17A)

- TFH-like markers (e.g. CXCL13, PDCD1, BCL6)
- Regulatory T cell markers (e.g. FOXP3, IL2RA)

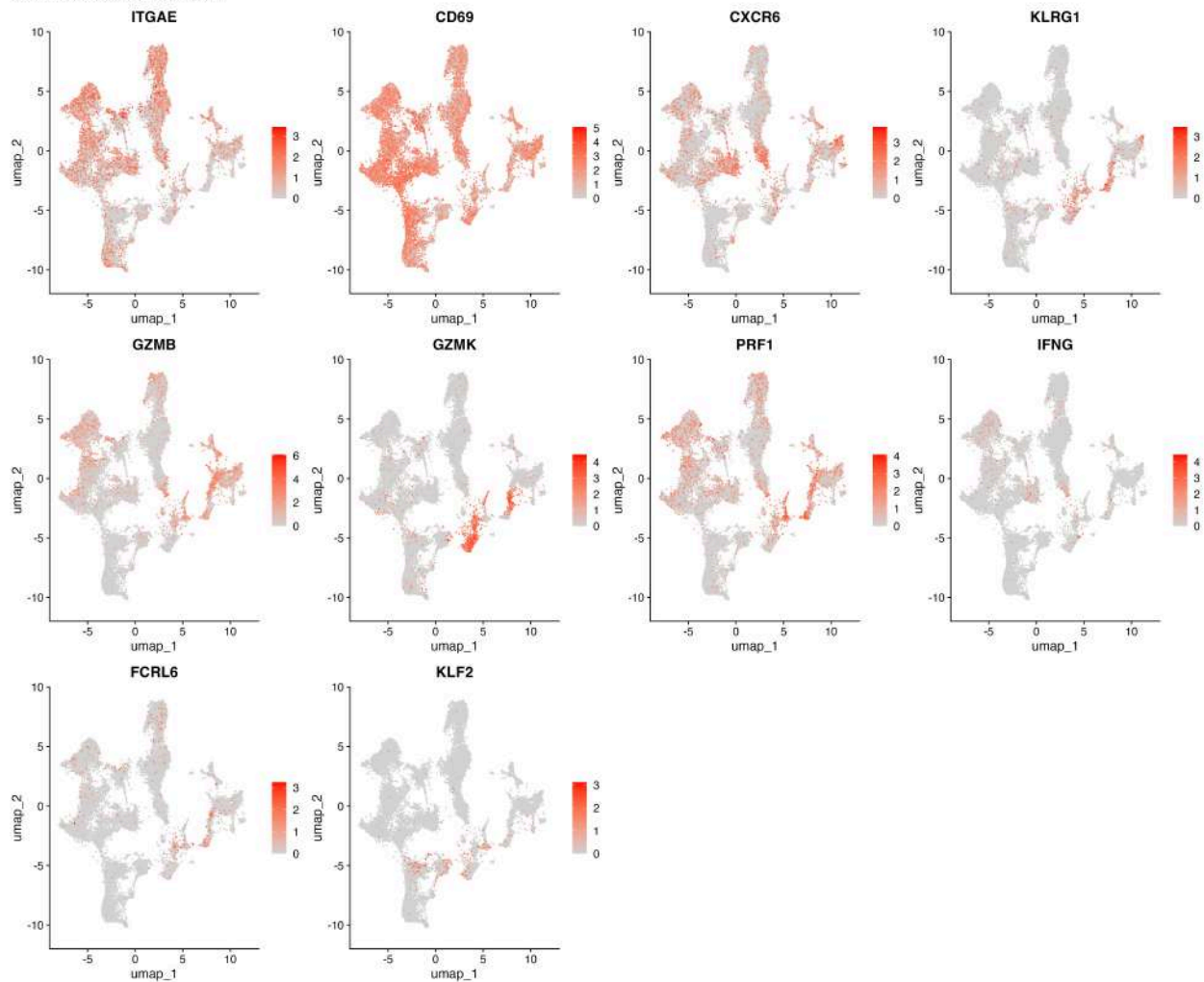
CD4 Markers

CD4 T cell functional programs



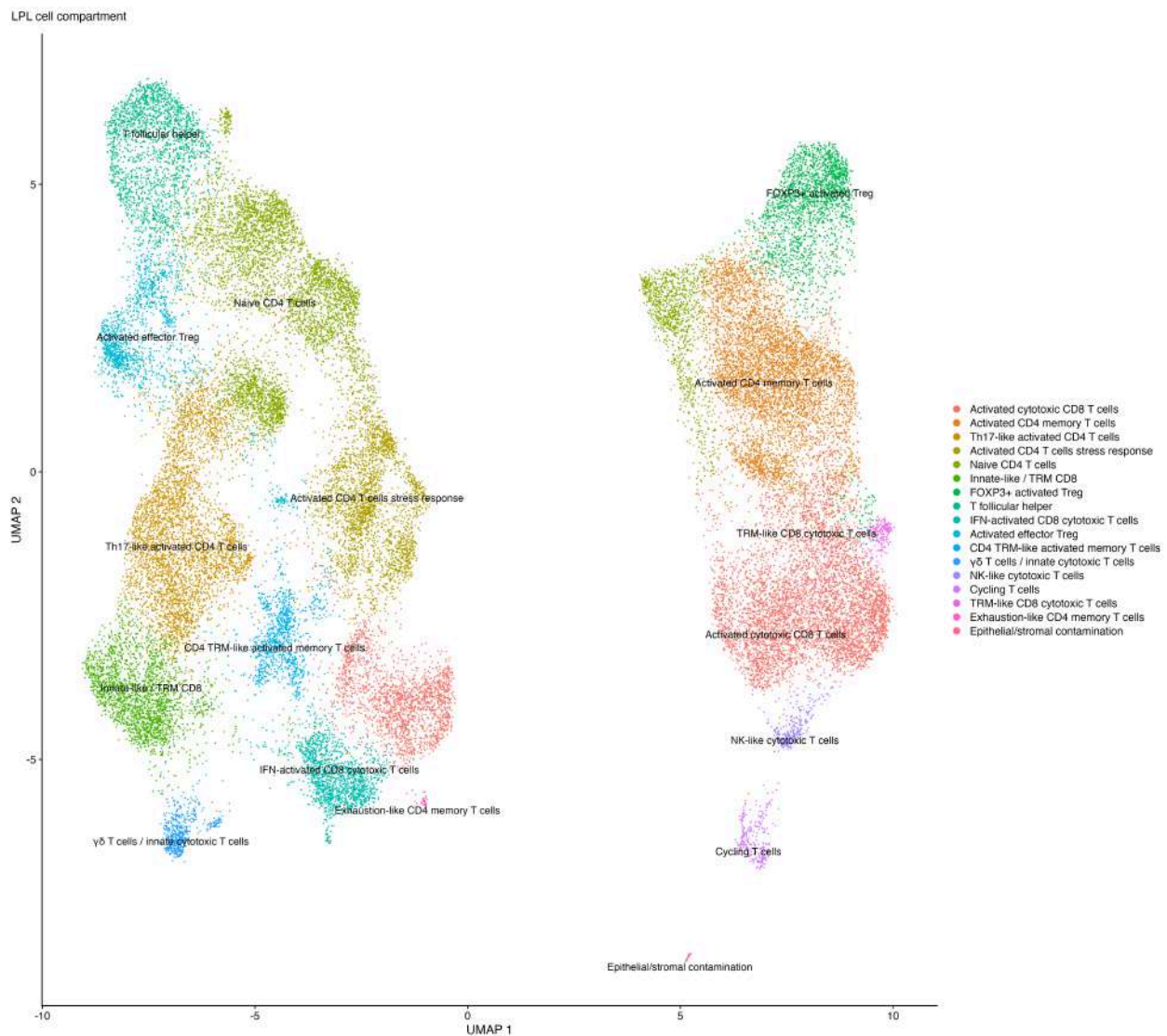
CD8 Markers

CD8 IEL residency & effector program



7. LPL Compartment Analysis and Reclustering

7.1 LPL transcriptional landscape



The LPL compartment shows higher transcriptional diversity and enrichment for adaptive immune regulatory and effector programs, including T helper, T follicular-like, regulatory T cell, and activated memory states.

7.2 LPL clustering and cell-type consolidation

Reclustering of LPL cells identified:

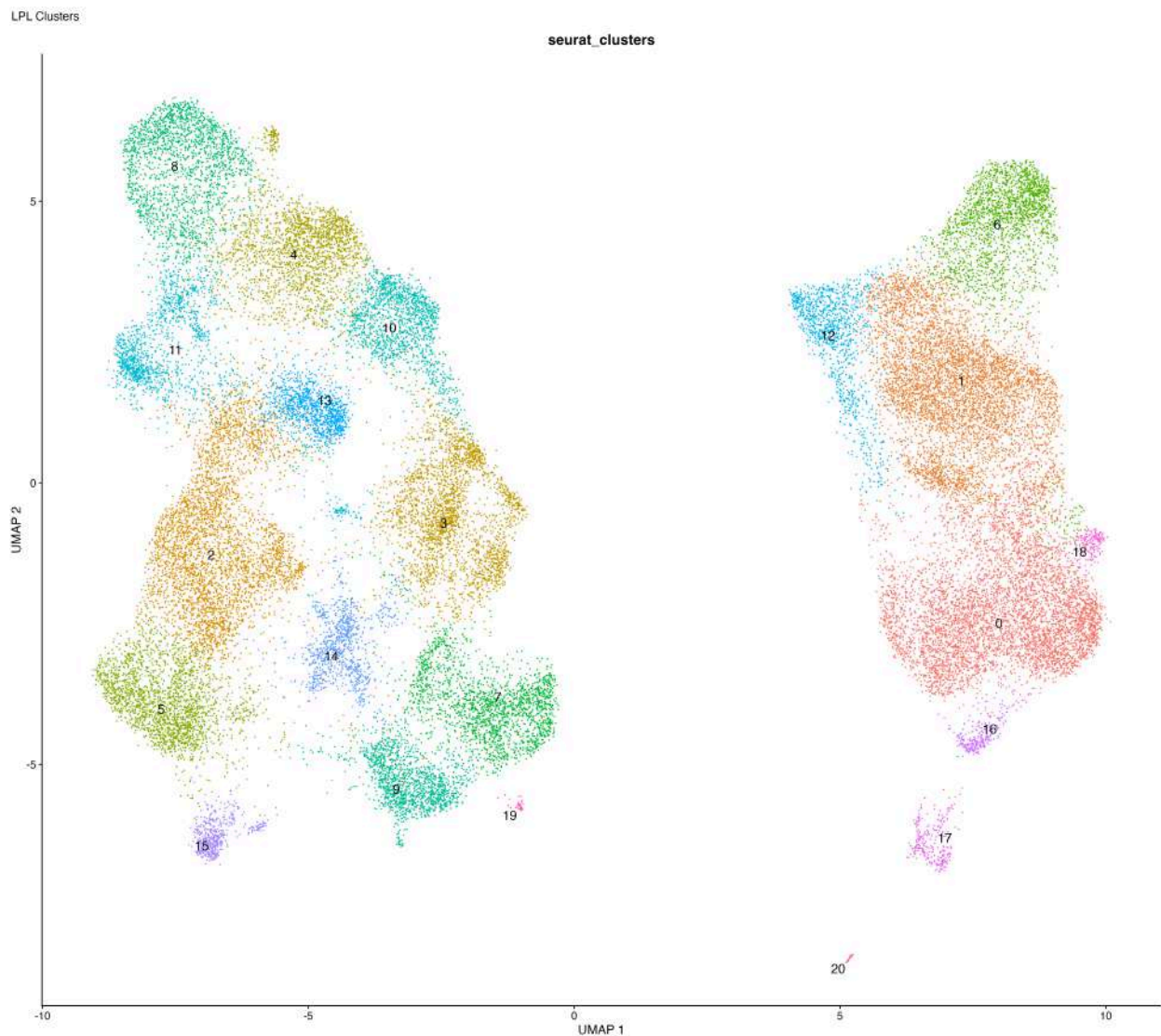
- 21 transcriptionally distinct clusters

These clusters were consolidated into:

- 17 biologically meaningful cell types

This reduction reflects partial redundancy among transcriptional clusters and convergence of activated/memory immune states into shared functional programs.

Clusters LPL Compartment



7.3 Key marker expression in LPL

LPL clusters were validated using canonical adaptive immune markers, capturing major transcriptional axes including:

- Helper T cell programs (IL7R, CCR7)
- T follicular-like signatures (CXCL13, PDCD1, BCL6)
- Regulatory T cell markers (FOXP3, CTLA4)
- TH17-like inflammatory signatures (IL17A, RORA)
- Cytotoxic and innate-like T cell programs (e.g. GZMB, NKG7)

CD4 Markers

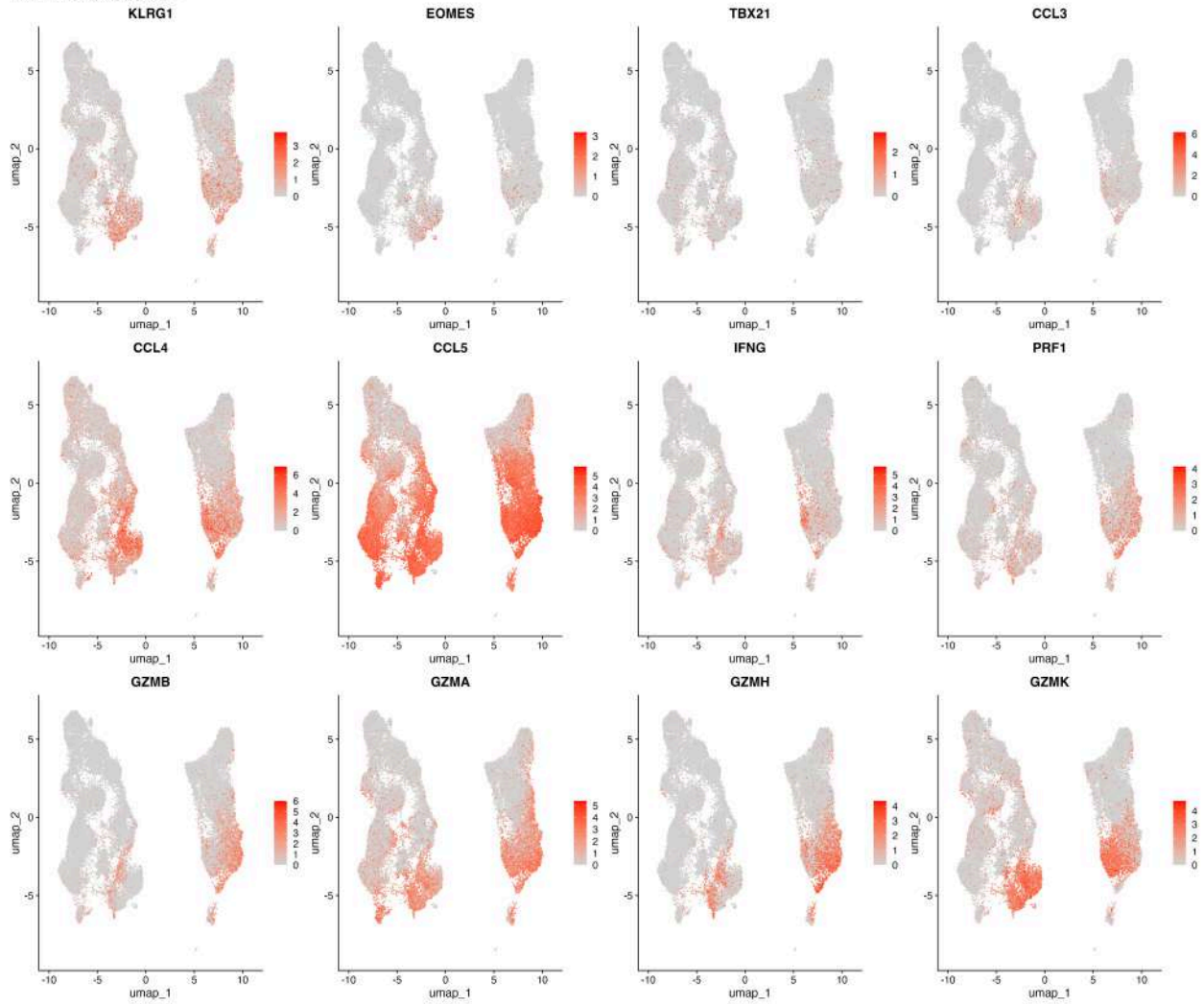
CD4 T cell functional programs



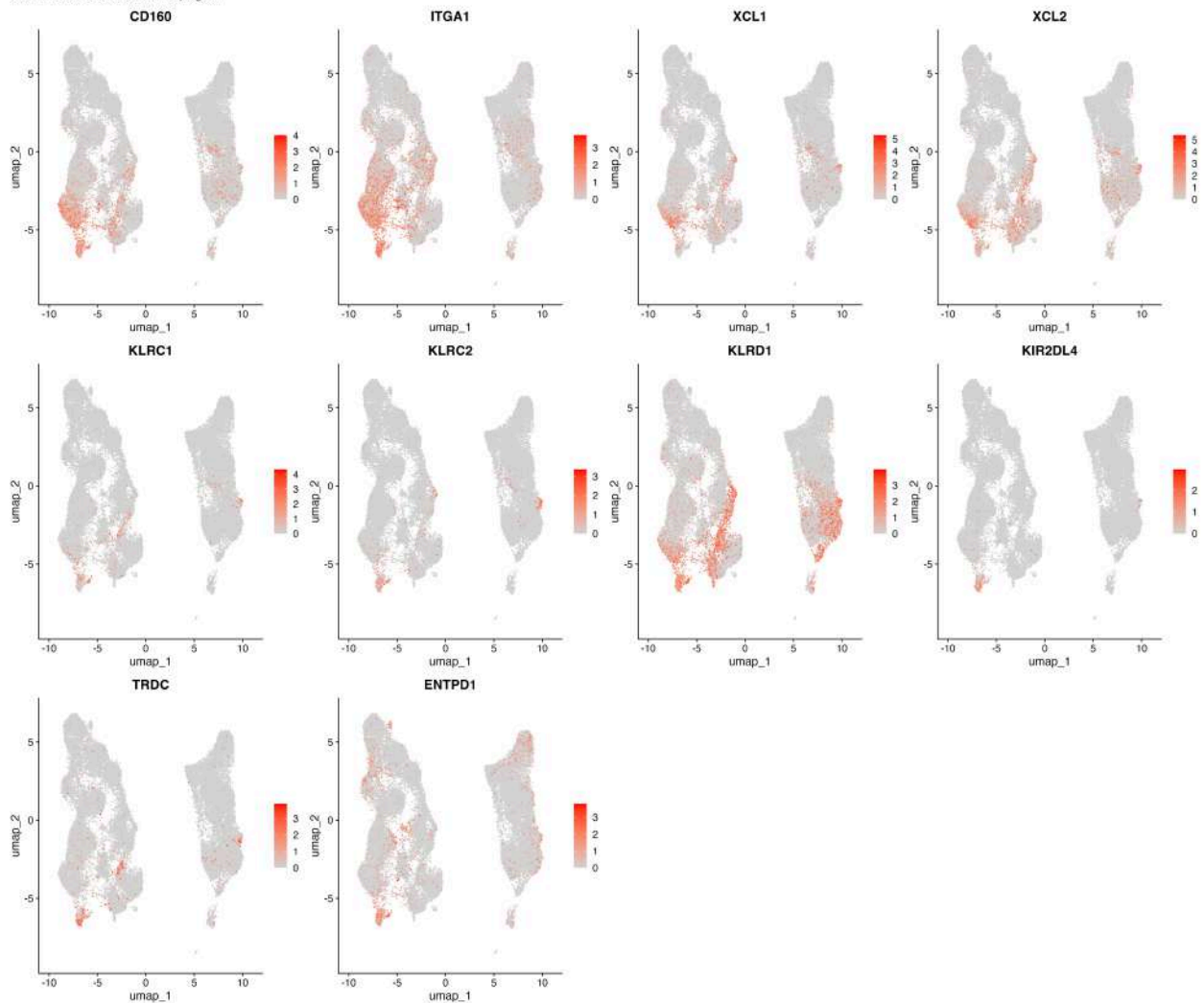
CD8 Markers

CD8 Effector Cells

CD8 LP effector cytotoxic program

*CD8 NK-like Cells*

CD8 LP tissue-associated NK-like program



8 Compartment-Specific Biological Interpretation

The compartmental decomposition of the global atlas reveals two distinct organizational principles of intestinal immunity:

IEL compartment

IEL cells exhibit high-resolution transcriptional fragmentation with near 1:1 mapping between clusters and cell types.

This reflects:

- strong epithelial interaction dependency
- coexistence of cytotoxic, $\gamma\delta$, and innate-like programs
- high cellular plasticity within the epithelial niche

LPL compartment

LPL cells show partial redundancy among transcriptional clusters, which were consolidated into fewer biologically meaningful cell types.

This reflects:

- convergence of CD4/CD8 memory and activated states
- presence of regulatory and TFH-like immune programs
- modular immune organization with shared transcriptional axes

9. Downstream Analytical Utility

The resulting compartment-specific objects support:

1. differential gene expression analysis within IEL and LPL
2. disease vs control comparisons per compartment
3. pathway enrichment and functional profiling
4. identification of compartment-specific disease signatures
5. trajectory or state-transition analysis within niches

10. Reproducibility

All analyses were performed within the same standardized computational framework used for the global analysis.

Environment:

- R 4.6.0
- Seurat 5.5.0
- Harmony 2.0.3
- SingleR 2.14.0
- Azimuth 0.5.1
- CellMarker 2.0 database

All intermediate Seurat objects are stored under:

objects/subsetting/

and all visual outputs under:

IEL Compartment: `results/plots/IEL_compartment/`

LPL Compartment: `results/plots/LPL_compartment/`

11. Limitations

Compartment-specific reclustering (IEL, LPL) was performed on RNA assay layers with a hybrid join state inherited from the global object: the `data` layer was joined across samples, while `counts` layers remained split by sample. As a consequence, variable feature selection followed a per-sample ranking procedure (Seurat v5 default behavior for split layers) rather than a fully pooled computation. This provides partial, unintentional mitigation of inter-sample technical variation, distinct from and weaker than formal batch integration. Downstream functional enrichment results (see `REPORT_GSEA`) showing biologically coherent, literature-consistent signatures (e.g., TH17 inflammatory activation, cytotoxic OXPHOS programs matching Jaeger et al. 2021) provide indirect support that cluster structure is not dominated by residual technical variation, though a formal sensitivity comparison (e.g., against a fully-joined or Harmony-integrated reanalysis) was not performed.

Subsetting introduces reduced cell numbers per analytical unit, which may affect:

- clustering stability in rare populations
- detection of low-frequency transcriptional states

However, this is mitigated by: - consensus annotation established at the global level prior to compartment-specific reclustering - retention of the full-resolution global object for cross-reference

12. Conclusion

The compartment-specific subsetting of IEL and LPL populations refines the global single-cell atlas into functionally and spatially coherent immune niches.

This step enables higher-resolution biological interpretation of intestinal immune architecture and provides the foundation for downstream disease-focused analyses within each compartment.